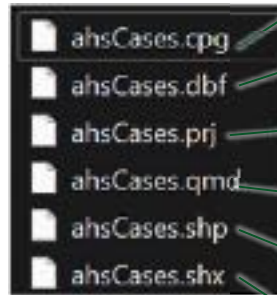


A vector shapefile



Specifies the character encoding used in the .dbf file to ensure correct text representation.

A dBASE database file that contains attribute data for the features in the .shp file (e.g., case ID, location, date, severity, etc.)

Stores information about the coordinate system and projection used in the shapefile.

QGIS metadata file used for styling or project-specific settings related to the dataset.

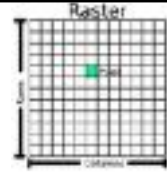
This is the core file that contains the geometry (points, lines, or polygons)

A companion file that stores an index to speed up the reading of geometric data.

Vector data is typically held in shapefiles, GeoJSON, KML or in Spatial databases like PostGIS

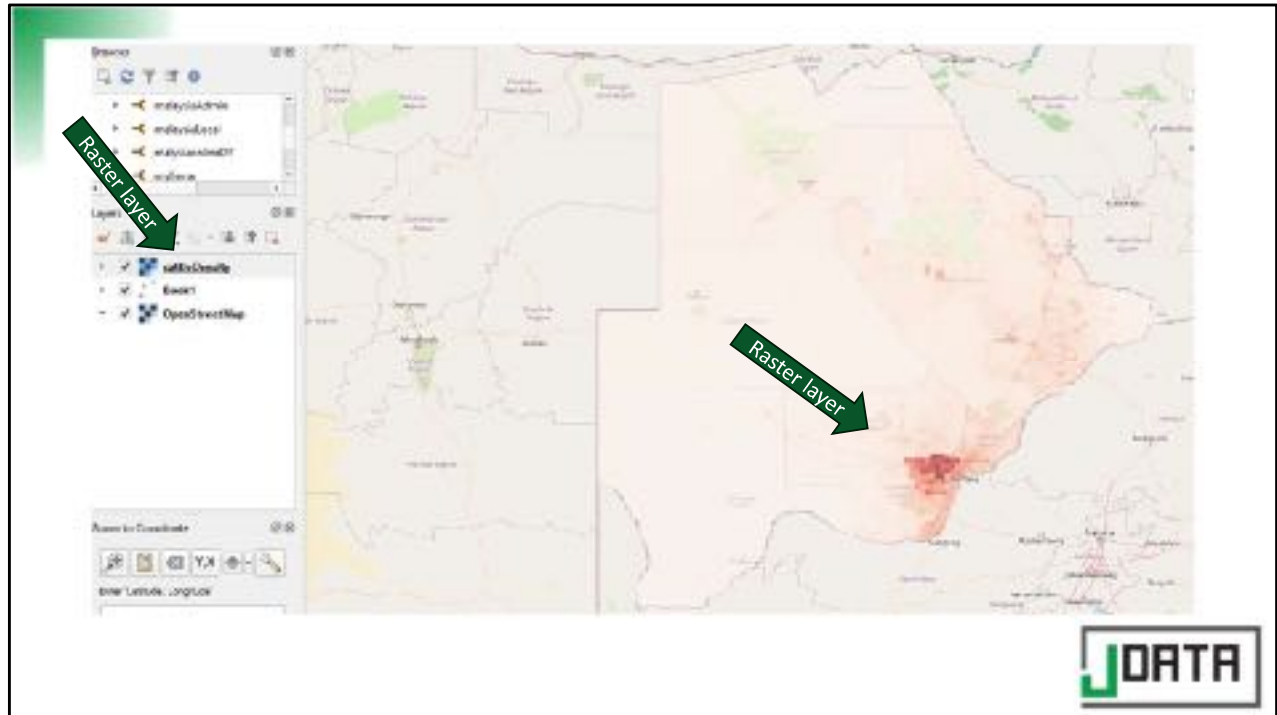


Raster data



- A raster data type is essentially a grid-based representation of a spatial area. Each cell in the grid holds a value that represents some attribute like elevation, temperature, or color.
 - Climate modeling
 - Elevation mapping
 - Land cover analysis
- GeoTIFF, JPEG, PNG
- Good for representing continuous data
- Easier and quicker to process computationally
- Less accurate for representing sharp boundaries or lines
- Large file sizes for high-resolution data





Raster resolution

- Each pixel represents a certain distance squared
 - High resolution versus Low resolution



Raster data is held in image file formats like GeoTIFF, JPEG



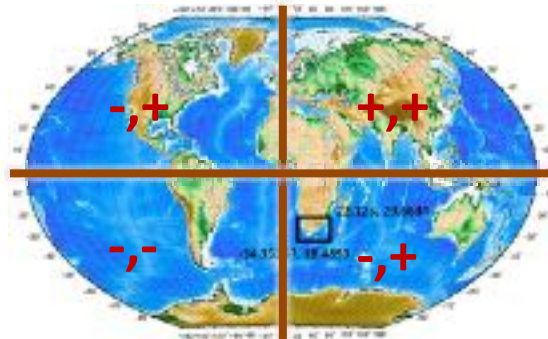
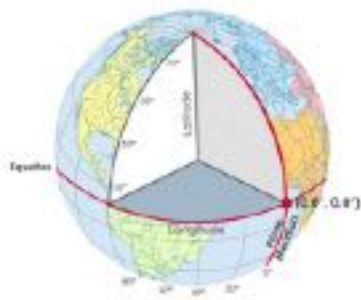
Coordinate Reference Systems

- **Map projections** transform (try to) the earth from its spherical shape (3D) to a planar shape (2D)
- A **coordinate reference system (CRS)** then defines how the two-dimensional, projected map in your GIS relates to real places on the earth.



Geographic coordinate systems

- degrees of latitude and longitude and sometimes also a height value to describe a location on the earth's surface



Lat	Long
South	East
Y	X



Degrees Minutes Seconds... Decimal Degrees

$60^{\circ} 35' 15''$

$15 \div 3,600 = 0.004167$

$35 \div 60 = 0.583333$

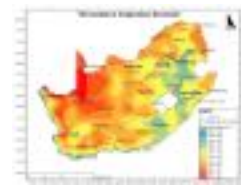
$+ 60$

60.5875



Spatial Analysis

- Buffers
 - Area within and beyond a distance from a feature
- Interpolation
 - Spatial interpolation is the process of using points with known values to estimate values at other unknown points



SECTION 2

- Introduction to qGIS



qGIS



- qGIS is the software component of GIS typically used for those wanting a open source option
- Install info
 - https://docs.qgis.org/3.40/en/docs/user_manual/introduction/getting_started.html#installing-qgis



Installing QGIS

- <https://qgis.org/en/site/forusers/download.html>



QGIS Interface - Menu bar



•**Project:** Operations like creating a new project, opening an existing project, saving, and exporting.

•**Edit:** Tools for editing features and attributes.

•**View:** Options for controlling the view of the map canvas, such as zooming and panning.

•**Layer:** Options for adding, removing, and managing layers.

•**Settings:** General settings, project properties, and customization.

•**Plugins:** Access to QGIS plugins.

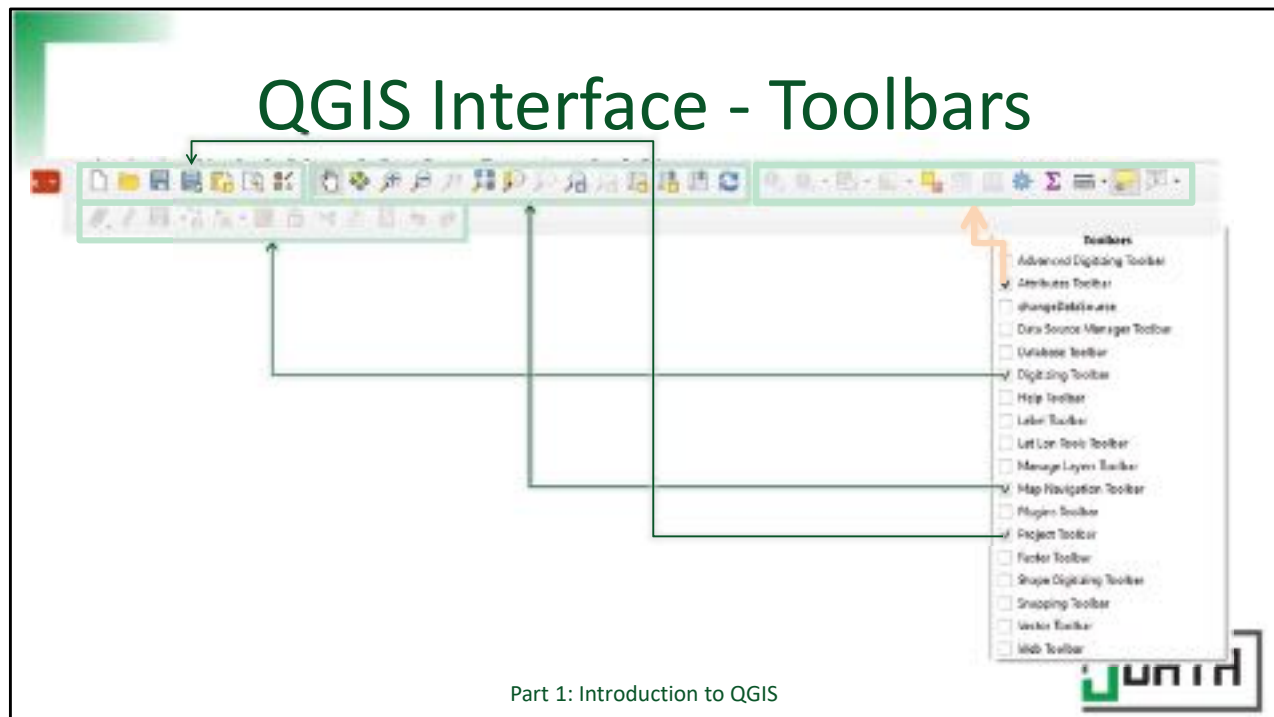
•**Raster:** Tools for raster data analysis and manipulation.

•**Vector:** Tools for vector data analysis and manipulation.

•**Web:** Tools for adding web maps and services.

•**Processing:** Opens the Processing Toolbox for various geoprocessing tasks.





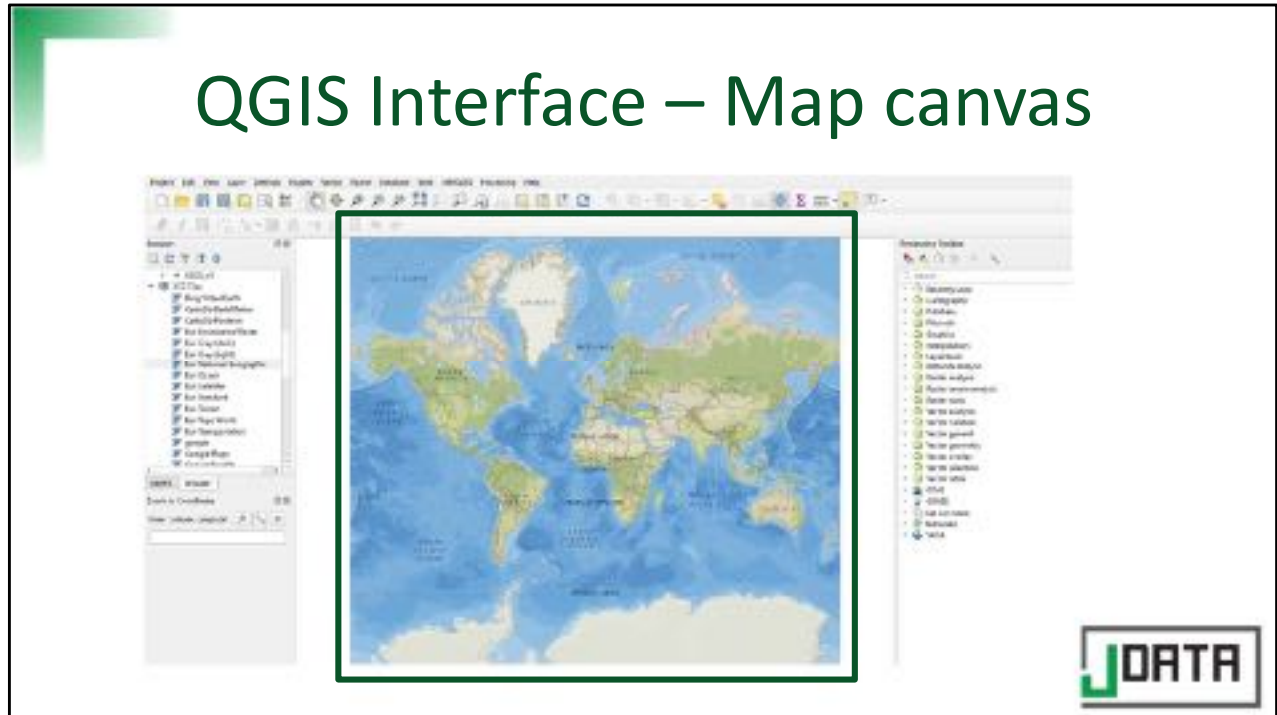
Attributes Toolbar: Tools for identifying features, measuring distances, and accessing the attribute table of a layer.

Digitizing Toolbar: Set of tools if you're working with vector data and need to create new geometries (points, lines, polygons) or edit existing ones.

Map Navigation Toolbar: Provides icons for zooming in/out, panning, and returning to the full extent of the map.

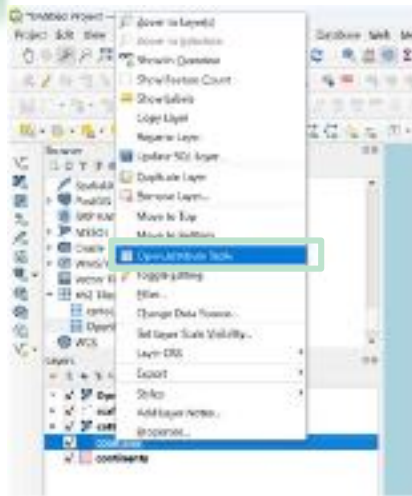
Project Toolbar: Contains icons for creating a new project, opening an existing one, saving the current project, and printing the map.

QGIS Interface – Map canvas



Map canvas: main area where your map layers are displayed. You can interact with the map here, performing tasks like zooming, panning, selecting features, amongst others

QGIS Interface – Attribute Table



A screenshot of the QGIS Attribute Table for the 'continents' layer. The table displays 20 rows of data, including continent names and their corresponding ISO codes.

ID	NAME	ISO3	ISO2	ISO1	ISO4
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9
10	10	10	10	10	10
11	11	11	11	11	11
12	12	12	12	12	12
13	13	13	13	13	13
14	14	14	14	14	14
15	15	15	15	15	15
16	16	16	16	16	16
17	17	17	17	17	17
18	18	18	18	18	18
19	19	19	19	19	19
20	20	20	20	20	20



QGIS Interface – Processing Toolbox

